

**Math 514, F13**  
**Exam 2**

Name: *Answer*  
SSN: xxx-xx-

**Instructions:**

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.

| Problem | Points | Score |
|---------|--------|-------|
| 1       | 20     |       |
| 2       | 15     |       |
| 3       | 20     |       |
| 4       | 15     |       |
| 5       | 15     |       |
| 6       | 20     |       |
| Total   | 100    |       |

**Good Luck !**

**Problem 1.** (20 points)

(a) The income stream  $\mathbf{a} = (1000, 1500, 2500, 4000)$  consists of payments made at the end of the next four years. Find the present value of  $\mathbf{a}$  for a nominal interest rate of 9% compounded continuously.

$$\begin{aligned} \gamma &= 0.09 \\ PV(\mathbf{a}) &= 1000 \cdot e^{-\gamma} + 1500 \cdot e^{-2\gamma} + 2500 \cdot e^{-3\gamma} + 4000 \cdot e^{-4\gamma} \\ &\approx 6865.99 \end{aligned}$$

(b) A loan for \$100,000 to be paid back over 180 months at a nominal yearly rate of 6% compounded monthly, what would be the monthly payment?

$$\begin{aligned} \frac{A}{1+\gamma} + \frac{A}{(1+\gamma)^2} + \dots + \frac{A}{(1+\gamma)^{180}} &= 100,000 \quad \text{where } \gamma = \frac{0.06}{12} = 0.005 \\ \text{let } \beta &= \frac{1}{1+\gamma} = \frac{1}{1.005}, \text{ then} \\ A\beta + A\beta^2 + \dots + A\beta^{180} &= 100,000 \\ A\beta \frac{1-\beta^{180}}{1-\beta} &= 100,000 \\ A &= 100,000 \cdot \frac{1-\beta}{\beta(1-\beta^{180})} \approx 843.86 \end{aligned}$$

**Problem 2.** (15 points) A certain investment costs \$3400. The investment produces income of \$1500 at the end of the first year and \$2500 at the end of the second year. Find the annual rate of return on the investment.

$$\begin{aligned} 3400 &= \frac{1500}{1+\gamma} + \frac{2500}{(1+\gamma)^2} \\ \text{let } \beta &= \frac{1}{1+\gamma}, \text{ we get} \\ 3400 &= 1500\beta + 2500\beta^2 \\ \Rightarrow 25\beta^2 + 15\beta - 34 &= 0 \\ \Rightarrow \beta &= \frac{-15 \pm \sqrt{15^2 - 4 \cdot 25 \cdot (-34)}}{2 \cdot 25} \approx 0.9042 \\ \Rightarrow \gamma &= \frac{1}{\beta} - 1 \approx 0.1060 \end{aligned}$$

**Problem 3.** (20 points) Assuming that the nominal interest rate is 4.8% per year, compounded continuously.

(a) A stock trades today at \$68.32, find a theoretical lower bound for the price of a call option with strike price  $K = \$67.50$  expiring two months from now.

By the put-call option parity formula,

$$S + P - C = Ke^{-rt}$$

$$C = S + P - Ke^{-rt}$$

$$\geq S - Ke^{-rt} \text{ (since } P \geq 0 \text{)}$$

$$= 68.32 - 67.50 e^{-0.048 \times \frac{1}{6}}$$

$$\approx 1.358$$

(b) Suppose that the no-arbitrage price of a call option (with strike price and time to expiration as in (a)) is \$1.95. Find the no-arbitrage price for a European put option with the same strike price and expiration date.

$$S + P - C = Ke^{-rt}$$

$$\Rightarrow P = Ke^{-rt} + C - S$$

$$= 67.50 e^{-0.048 \times \frac{1}{6}} + 1.95 - 68.32$$

$$\approx 0.592$$

**Problem 4.** (15 points) Let  $C$  be the price of a call option to purchase a security whose present price is  $S$ , argue that  $C \leq S$ .

If  $C > S$ , we always obtain an arbitrage by simultaneously selling a call option and buying a security.

{ Initial yield:  $C - S > 0$

{ Payoff at the expiration time: either holding of the stock or  $K$ , the strike price ( $> 0$ ).

**Problem 5.** (15 points) An experiment can result in any of the outcomes 1, 2, or 3. If there are two different wagers with

$$r_1(1) = 2, r_1(2) = 4, r_1(3) = -3,$$

$$r_2(1) = 4, r_2(2) = 6, r_2(3) = -5,$$

Is an arbitrage possible?

We need check if there is a probability vector  $(P_1, P_2, P_3)$  such that

$$\begin{cases} \sum_{j=1}^3 P_j r_1(j) = 0 \\ \sum_{j=1}^3 P_j r_2(j) = 0 \\ \sum_{j=1}^3 P_j = 1 \end{cases} \Rightarrow \begin{cases} 2P_1 + 4P_2 - 3P_3 = 0 \\ 4P_1 + 6P_2 - 5P_3 = 0 \\ P_1 + P_2 + P_3 = 1 \end{cases} \Rightarrow \begin{cases} P_1 = \frac{1}{4} \\ P_2 = \frac{1}{4} \\ P_3 = \frac{1}{2} \end{cases}$$

Thus no arbitrage is possible according to the arbitrage theorem.

**Problem 6.** (20 points) Suppose that the price of a security either goes up by a factor of 2 or goes down by a factor of  $1/2$  in one period (i.e.,  $u = 2$  and  $d = 1/2$ ). We also assume the periodically compounded nominal interest rate is  $r = 10\%$  and the initial price of the security is 100.

(a) Compute the risk-neutral probabilities  $(p^*, 1 - p^*)$ .

$$p^* = \frac{1+r-d}{u-d} = \frac{1+0.1-0.5}{2-0.5}$$

$$= \frac{0.6}{1.5} = 0.4$$

$$1-p^* = 1-0.4 = 0.6$$

(b) Determine the no-arbitrage cost of a European put option to sell the security at the end of one period for a strike price of 150.

Assume we buy a put option at time 0.

$$\text{return at time 1} = \begin{cases} -P & \text{if the security price is } 100 \times 2 = 200 \\ 100(1+r)^{-1} - P & \text{if the security price is } 100 \cdot \frac{1}{2} = 50. \end{cases}$$

where  $P$  denote the cost of the put option.

$$\begin{aligned} E[\text{return}] &= -P \cdot P^* + (100(1+r)^{-1} - P) \cdot (1 - P^*) \\ &= -P + \frac{100}{1+r} \cdot (1 - P^*) \end{aligned}$$

$$\begin{aligned} &= 0 \\ &\quad \mapsto \text{by the Arbitrage Theorem} \end{aligned}$$

$$\begin{aligned} \Rightarrow P &= \frac{100}{1+r} (1 - P^*) \\ &= \frac{100}{1+0.1} \cdot 0.6 \\ &\approx 54.55 \end{aligned}$$